

Sparse Cross-Document Contextualization for Persistent Native Memory

An Oracle Headroom Study beyond Packed RAG

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Draft — September 2026

Abstract

Retrieval-augmented generation (RAG) normally serializes selected documents into one causal token sequence, so later documents are contextualized by earlier documents before the query is decoded. Progressive Retrieval Attention (PRA) instead allows selected resources to remain independently addressable and to persist as native key/value memory. Prior work established that contiguous PRA transport can preserve model behavior exactly, that pre-RoPE keys can be rebound to request-specific packed positions, and that independently stored resources closely reproduce a packed computation when document-to-document attention is disabled. The remaining quality gap is therefore primarily a cross-document-contextualization problem rather than a position-transport problem.

This paper asks whether the useful portion of packed-RAG cross-document interaction is sparse. We instrument the unchanged host-model attention path, extract an oracle graph over physical layer/head/token edges, and replay the same frozen retrieval selection under exact sparse masks. Candidate identities, record order, prompt semantics, model revision, and generation settings remain fixed. Compressed graphs retain only causal cross-record edges; immutable selection receipts and graph digests tie every intervention to its source.

The inception experiment evaluates edge-count budgets from 0 to 100% and cumulative teacher-mass targets from 50 to 99% on MultiHop-RAG. It also freezes disjoint 2,000/150/150 train/validation/test identities for a possible learned selector. On a ten-question mechanism cohort, 0.1% of physical edges reached .1488 F1 versus .1490 for packed RAG and .1431 for independent PRA, but this is 97.3% recovery of a small 0.0059 gap and the cohort’s packed official score was only .600. We therefore run a powered 150-question frozen test comparing raw attention with document-pair leave-one-group-out NLL, first-step JS, and a finite-difference NLL-times-attention proxy. Pair NLL at 1% obtains .1243 F1 and .3933 official score versus .1119 and .3333 for packed RAG; paired differences are +.0124 F1 and +.0600 official, but both 95% intervals include zero. Every frontier remains non-monotonic and the absolute .19/.67 gate is missed. The result supports sparse causal headroom and shows that task-aware rankings improve on raw magnitude, but it does not justify learned selector training.

1 Introduction

RAG systems solve a retrieval problem and a context-construction problem. Retrieval selects useful evidence. Context construction then decides how that evidence is presented to the language model. The common implementation packs retrieved text into a causal sequence,

$$D_1 D_2 \cdots D_m Q,$$

and freshly prefills the whole sequence. This creates a dense and order-dependent computation graph: D_2 can attend to D_1 , D_3 can attend to both D_1 and D_2 , and the final query can attend to all preceding states.

PRA exposes a different design point. Each D_i can remain a persistent addressable record with reusable model-native K/V. The query can later select and materialize those records without rebuilding all source states. The advantage is straightforward when a selection repeats or when records recur across changing queries. The harder question is composition: independent records do not contain the cross-document causal history introduced by ordinary packed RAG.

Paper 3.2 localized this difference. Contiguous selected-context transport was behaviorally exact across multiple model scales and families. Pre-RoPE request-time rebinding reproduced the positional geometry of a packed request, and independently encoded records closely tracked a packed control in which document-to-document attention was disabled. In the powered natural comparison, however, packed RAG remained stronger than independent PRA. A small trained residual recovered part of the gap but did not establish a qualified solution. Fixed sparse masks were non-monotonic, while one very small boundary interaction budget approached packed F1 on some metrics.

These results motivate a narrower question:

How much cross-document interaction is actually necessary, and can the necessary interactions be predicted and executed sparsely at request time?

We therefore separate three quantities:

1. **persistent source state**, stored once as reusable PRA records;
2. **request-specific interaction selection**, which determines which records/tokens/layers need contextualization;
3. **request-specific interaction execution**, which applies a small amount of the frozen base model’s own computation.

The intended architecture is

persistent PRA K/V $\rightarrow$ sparse interaction selector
 $\rightarrow$ small request-local cross-record computation $\rightarrow Q$.

Contributions. This inception study makes four contributions:

1. A host-preserving attention observer and sparse-mask replay path that leaves base-model parameters and persistent records unchanged.
2. A compressed physical-edge graph and immutable interaction-plan receipt spanning layers, heads, record pairs, and token positions.
3. A prespecified oracle sparsification study over edge-count and cumulative-attention-mass budgets, with task and distribution metrics.
4. A leakage-safe data split and an explicit learned pair-selector design that remains locked unless oracle evidence justifies training.

2 Related Work

RAG and retrieval-conditioned language models place external evidence in the model context or retrieve it during generation [5, 4, 3]. Sparse-attention systems reduce long-sequence cost through fixed, random, global, or learned access patterns [1, 10, 7]. Other work retrieves external states at attention time or restricts K/V traffic [2, 8, 6]. These approaches motivate sparse access, but the present question is different: the retrieved records are already selected and persist as independently encoded native memory. We ask which request-specific cross-record interactions must be restored to recover the behavior of a conventional packed prefix. CacheBlend also studies reuse when separately cached chunks need request-time fusion [9]; our unit of intervention is an auditable physical attention edge under a frozen RAG selection.

3 Relationship to Prior PRA Work

3.1 Paper 1.5: request-time RoPE binding

Paper 1.5 separated coordinate correctness from causal contextualization and showed that, for RoPE models,

$$K_{\text{post}}(p) = R_p K_{\text{pre}}.$$

Pre-RoPE storage therefore allows a persistent key to be rebound to a request-specific effective position without re-encoding the source.

Paper 3.3 assumes that positional contract and does not reopen the positional transport question except as a runtime qualification requirement.

3.2 Paper 3.0: materialization is a computational choice

Paper 3.0 showed that more materialized K/V is not automatically better and that useful detail depends on the query, evidence distribution, and consumer layer. This motivates a query-conditioned interaction policy rather than a fixed global cross-document mask.

3.3 Paper 3.2: causal localization of the composition gap

Paper 3.2 supplies the direct starting point. With a strong frozen selector, packed RAG reached approximately .199 token F1 and .700 official score on the final held-out comparison, while independent PRA reached approximately .135 and .433. A rank-8 request-local residual raised PRA to approximately .154 F1 and .533 official score but left substantial headroom and an uncertain five-seed F1 effect.

More importantly, Paper 3.2 compared

$$A = \text{full causal packed RAG},$$

$$B = \text{packed RAG with document-to-document attention disabled},$$

and

$$C = \text{independent pre-RoPE PRA records rebound to the same positions}.$$

B and C were behaviorally close, while a shape-matched storage/rebinding control was exact across the tested precision modes. The principal semantic difference is therefore cross-document causal interaction.

Paper 3.3 does not attempt to make independent records reproduce every dense packed state. It asks which subset of those interactions is needed for task quality.

4 Problem Formulation

Let the selected evidence consist of m persistent records

$$\mathcal{D} = \{D_1, \dots, D_m\},$$

with total token count T . Let $\mathcal{E}_{\text{full}}$ denote all document-to-document attention edges allowed by ordinary packed causal RAG.

We seek a request-specific sparse edge set

$$\mathcal{E}_q \subset \mathcal{E}_{\text{full}}$$

such that

$$|\mathcal{E}_q| \ll |\mathcal{E}_{\text{full}}|$$

while preserving task utility:

$$U(\mathcal{D}, Q, \mathcal{E}_q) \approx U(\mathcal{D}, Q, \mathcal{E}_{\text{full}}).$$

The system must additionally preserve most persistent native state:

$$C_{\text{new}}(\mathcal{E}_q) \ll C_{\text{prefill}}(\mathcal{D}).$$

Our primary quality target is to recover at least 80% of the independent-to-packed RAG gap under a request-specific interaction budget of at most 10% of dense cross-document interaction.

5 Oracle Sparse Interaction

5.1 Teacher graph extraction

We instrument packed RAG with frozen retrieval, reranking, selected records, order, token budget, model, and generation settings. For every causal document-to-document attention interaction we record sufficient metadata to construct a sparse teacher graph:

- layer and head;
- source and target record/chunk;
- source and target token positions;
- attention mass;
- distance to document boundaries;
- retrieval rank and score.

Let E be the set of causal token pairs whose source and target belong to different records. The stored score array has shape $[L, H, |E|]$, not $[L, H, T, T]$. Source and target token indices and record identities are stored once, so within-record and causally impossible entries never enter the artifact. For each layer, the observer computes probabilities on a diagnostic side path; the hidden-state update still uses MLX’s host scaled-dot-product-attention kernel. A parity receipt checks the observer path against ordinary host execution before the graph is interpreted.

5.2 Oracle budgets

We first retain the highest-attention edges at budgets

$$0, 0.01, 0.05, 0.1, 0.5, 1, 2, 5, 10, 100\%.$$

A complementary cumulative-mass condition retains the smallest edge set covering

$$50, 75, 90, 95, 99\%$$

of cross-document attention mass.

For replay, a plan contains a Boolean array of shape $[L, H, |E|]$. At layer ℓ , selected physical edges are added to a packed mask that otherwise blocks cross-record attention. The frozen host model then executes the modified mask. The 100% plan must reproduce the packed teacher exactly; the 0% plan must reproduce the blocked packed control. These endpoint checks distinguish a sparsification result from an instrumentation or transport error.

5.3 Interventional ranking targets

Attention magnitude is observational: a large coefficient shows that an interaction occurred, not that it improved the answer. We therefore add a leave-one-group-out utility for a group g ,

$$u_{\text{NLL}}(g) = \mathcal{L}_{\text{answer}}(M_{\setminus g}) - \mathcal{L}_{\text{answer}}(M),$$

where M is the packed teacher and $M_{\setminus g}$ removes only the group’s cross-document edges. A positive value means that removing the group raises gold-answer NLL. We also measure first-step Jensen–Shannon divergence, $u_{\text{JS}}(g)$, between $M_{\setminus g}$ and M .

The hierarchy is document pair $\rightarrow$ layer $\rightarrow$ layer/head $\rightarrow$ token edge. At each tested level, groups are ordered by measured utility and physical edges within a group are refined by teacher attention. A second pair-level proxy ranks by

$$\max(0, u_{\text{NLL}}(g)) a_e,$$

where a_e is the teacher attention of edge e in group g . This is a finite-difference-times-attention diagnostic, not a gradient attribution. These rankings are compared with raw attention at matched physical-edge budgets 0, .01, .05, .1, .5, and 1%; 100% remains an endpoint parity check.

Every natural question is sampled only after restricting the dataset to a named frozen Paper 3.3 split. Atomic per-question checkpoints bind the dataset, model, reranker, split digest, ranking targets, and budgets, so a resumed run cannot mix configurations. A ten-question validation diagnostic localizes the .1% plans by layer, head, and document pair and selects ranking targets. The powered confirmation then requires at least 100 previously untouched test questions. Learned selection remains locked unless an interventional test ranking meets the absolute quality gate and its F1 frontier is non-decreasing through 1%.

5.4 Oracle gate

The learned-selector program proceeds only if a sparse oracle can approach packed-RAG quality. The initial gate is

$$\text{F1} \geq .19, \quad \text{official} \geq .67,$$

or statistical indistinguishability from packed RAG while using no more than 5% of dense cross-document interactions.

6 Where Useful Interactions Live

We analyze oracle-selected interactions by:

- transformer layer;
- attention head;
- source/target retrieval rank;
- document pair;
- chunk pair;
- token boundary distance;
- query overlap;
- entity/token overlap;
- gist similarity.

This analysis determines the appropriate routing granularity. If useful interactions are largely pair-level, we avoid token-level prediction. If they concentrate in a few layers or heads, request-local compute can be reduced further.

7 Learned Interaction Selection

This section specifies the next-stage design; it is not evaluated in the present paper. Training is deliberately locked by the failed inception gate.

7.1 Pair selector

The first learned policy predicts interaction importance for selected pair (D_i, D_j) :

$$s_{ij} = f_{\theta}(g_q, g_i, g_j, r_i, r_j, \phi_{ij}),$$

where g_q is a query gist, g_i, g_j are persistent record/chunk gists, r_i, r_j are retrieval/reranker features, and ϕ_{ij} contains pairwise semantic features.

The pair selector is trained against oracle pair importance while controlling a request-specific interaction budget.

7.2 Layer/head selector

If the oracle is concentrated, we extend the policy to

$$s_{ij\ell h}$$

or a factored approximation.

The paper prefers the simplest selector that reaches the quality/economics gate.

8 Sparse Request-Local Contextualization

Synthetic parameter-free K/V append failed in Paper 3.2. The present paper therefore prioritizes on-manifold computation using the frozen base transformer’s own projections and attention operations.

Candidate mechanisms are:

Sparse packed-mask composition. Use selected interactions in a request-local composition pass as the first oracle/runtime control.

Boundary interaction. Permit selected real boundary tokens to interact across selected record pairs.

Selected-token interaction. Use a learned token gate within selected record pairs.

Selected-layer propagation. Apply interaction only at selected layers, then propagate corrected request states through the remaining decoder.

Persistent record K/V remains immutable. All corrections are request-local.

9 Retrieval-Guided Cross-Document Expansion

Physical-edge selection asks which neural interactions to replay. A cheaper alternative asks an earlier question: whether one selected record identifies a small, previously unselected interval in another selected record. Let $S_i \subset D_i$ be the frozen first-stage span for record D_i . For every allowed ordered pair $i \neq j$, we issue either

$$S_i \longrightarrow \text{search}(D_j) \quad \text{or} \quad (Q, S_i) \longrightarrow \text{search}(D_j).$$

The first form measures selected-span evidence alone; the second uses the user query to suppress associations unrelated to the current reasoning path. Search never escapes the records authorized by the frozen first-stage receipt.

We implement exact chunk BM25, frozen dense similarity, reciprocal-rank fusion, an entity/keyterm channel, and a support-span oracle. Powered dense runs use BAAI BGE-base-en-v1.5 with its query and document encodings kept distinct; the dependency-free signed-hash encoder is an SDK fallback and is not labeled as a semantic result. Dense vectors for authorized peer chunks, the request, and selected source spans are prepared once per request. We report that preparation separately from scorer latency so dense and fusion rows cannot inherit order-dependent warm-cache advantages. For lexical rank $r_{\text{lex}}(c)$ and dense rank $r_{\text{dense}}(c)$, the primary hybrid score is

$$s_{\text{RRF}}(c) = \frac{1}{60 + r_{\text{lex}}(c)} + \frac{1}{60 + r_{\text{dense}}(c)}.$$

RRF avoids calibrating unlike raw scores. A normalized weighted fusion is kept as a secondary research arm. We also compare symmetric all-pairs expansion, higher-to-lower and lower-to-higher retrieval rank, a top-record hub, and pairs that pass a query-relevance threshold.

The runtime, rather than the search policy, owns safety and accounting. It rejects unauthorized targets, removes overlap with the initial selection and other expansions in logical source coordinates, clips the final interval to a global token ceiling, and distinguishes resident from newly encoded native K/V. For candidate set C and admitted source intervals A , it records

$$B_{\text{cross}} \in \{32, 64, 128, 256, 512\}, \quad k_{\text{pair}} \in \{1, 2, 4, 8\},$$

Table 1: Best bounded cross-document expansion point per channel on the 150-question validation split. Gains are absolute percentage points over the same frozen first-stage selection. “Q” indicates query conditioning.

Channel	Q	Span gain	Chunk gain	Extra K/V	Distractor	Search ms
Lexical	yes	2.67	3.71	12.66%	90.00%	337.8
Dense	no	2.94	4.27	12.66%	86.00%	258.7
RRF hybrid	yes	2.44	3.79	12.66%	88.00%	344.7
Weighted hybrid	yes	1.83	4.35	12.66%	87.33%	340.5
Entity/keyterm	yes	1.72	2.71	12.66%	92.67%	337.0
Oracle	yes	23.17	30.55	19.65%	0.00%	338.2

together with requested, deduplicated, reused, and newly materialized tokens. The hash-addressed `CrossDocumentExpansionReceipt` contains source and target identities, offsets, channel scores, ranks, keyterms, policy revision, and failure labels, but not raw source text.

Expansion and neural interaction remain separate interventions. *Expansion only* appends selected native source intervals without cross-record SA. *Pair SA* admits base-model attention only between retrieval-linked record pairs. *Boundary SA* further limits each linked pair to a source suffix and target prefix. Finally, a diagnostic physical-edge arm ranks teacher edges only inside retrieval-linked pairs. These masks execute through the unchanged host attention implementation, so their edge fractions describe logical sparsity, not kernel acceleration.

The public abstraction is fail-closed. Parameter-free modes have stable configuration and plugin contracts, while oracle and weighted modes require an explicit experiment flag. Runtime authorization, deduplication, budget enforcement, and receipt generation also wrap custom plugins. No mode becomes an SDK default from a smoke result: qualification requires a powered frozen cohort, five-seed uncertainty, at least two model sizes, bounded quality and latency, and preferably cross-family replication.

The first-stage selection cache stores tokenizer-independent source intervals and checks the candidate receipt, selector identity, token ceiling, and record limit before replay. Model tokenization occurs only during native encoding. An interprocess receipt check during this work exposed two defects in the historical BM25-v1 implementation: corpus term frequency had been accumulated where document occurrence was required, and floating sums traversed an unordered query-term set. BM25-v2 counts each term once per document and sorts query terms before accumulation. We invalidated and regenerated all preliminary expansion results after this correction; every new receipt carries the v2 revision. Earlier physical-edge results remain tied to their pinned historical commit and are not silently relabeled as v2 comparisons.

9.1 Frozen validation frontier

We evaluate 221 policy configurations on 150 frozen validation questions. The grid crosses five parameter-free channels and an oracle with selected-only or query-conditioned probes, $k \in \{1, 2, 4, 8\}$, and extra-native-token ceilings from 32 to 512. Table 1 reports the best point per channel subject to an extra-native fraction of at most 20%. These are validation choices, not test results. All non-oracle optima use a 64-token ceiling, which is 12.66% of the initial 506.2-token native selection on average.

Dense selected-only is the predeclared test policy: symmetric chunk search, $k = 1$, and a 64-token global ceiling. It raises supporting-span coverage from 41.33% to 44.28% and gold-chunk recall from 38.57% to 42.85%. Five repeated paired bootstrap calculations place the span gain in a conservative 95% interval of [1.39, 4.72] points and the chunk-recall gain in [2.51, 6.22] points. Answer-string

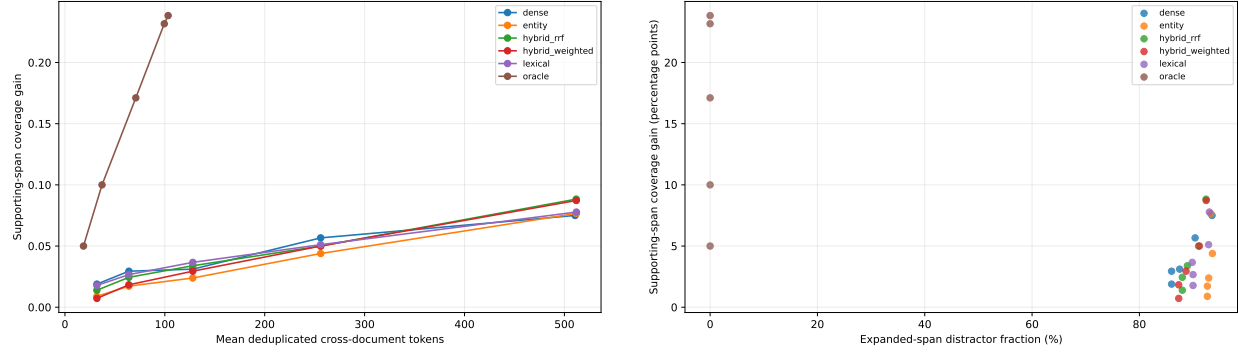


Figure 1: Validation expansion frontier. Left: support-span gain against additional native K/V. Right: gain against the fraction of admitted tokens outside annotated support. Oracle points expose substantial unresolved selection headroom.

availability rises by 2.67 points, with interval [0.67, 5.33].

The positive retrieval effect is not yet an efficiency or answer-quality win. The selected policy requests 1,521.1 cross tokens per question before overlap removal and global clipping, but admits only 64.0: a 23.77-fold request-to-materialization amplification. Moreover, 86% of admitted tokens do not overlap annotated support. The compact oracle gains 23.17 span points with 99.4 extra tokens and no distractors; dense expansion therefore recovers only 12.71% of oracle span headroom. Figure 1 makes both the bounded gain and the unresolved selectivity gap visible. Semantic vector preparation adds 358.7 ms per request in this exhaustive research implementation and is reported separately from the 258.7 ms dense search.

10 Training Objective

The base model remains frozen initially.

We optimize

$$\mathcal{L} = \lambda_{\text{task}} \mathcal{L}_{\text{answer}} + \lambda_{\text{seq}} KL(P_{\text{sparse}} \| P_{\text{packed}}) + \lambda_{\text{sel}} \mathcal{L}_{\text{selector}} + \lambda_{\text{cost}} C_{\text{interaction}}.$$

Task quality receives the dominant weight. Global K/V reconstruction is not a primary objective because Paper 3.2 showed that state proximity, answer NLL, and generated-answer quality need not agree.

11 Order Robustness

Packed RAG changes its causal graph when selected documents are permuted. PRA records offer the possibility of more stable set-like evidence composition.

We evaluate canonical, reverse, and random permutations under:

- packed RAG;
- independent PRA;
- oracle sparse composition;
- learned sparse composition.

We report task quality together with mean pairwise JS, unique outputs per question, and F1/NLL variance. Lower order sensitivity is considered a benefit only at matched or improved task quality.

12 Experimental Protocol

12.1 Dataset

MultiHop-RAG is the first mandatory dataset because Paper 3.2 already provides frozen retrieval, canonical records, support labels, and powered natural baselines.

Additional datasets are introduced only after the mechanism passes the primary gate.

12.2 Selection

Use the strongest qualified matched selector from Paper 3.2:

BM25 $\rightarrow$ BGE-v2-M3.

Candidate and selection receipts are frozen before composition conditions run.

12.3 Training split

The frozen split uses:

- 2,000 training questions;
- 150 validation questions;
- 150 fixed held-out test questions evaluated with seeds 11, 23, 37, 71, and 101.

The split seed is 3301. All 30 identities declared by the final Paper 3.2 rank-8 residual manifest are excluded from every Paper 3.3 split, rather than only from training. The dataset revision, corpus revision, exact identities, legacy manifest, and split digest are persisted. Oracle smoke questions do not train or select parameters.

12.4 Model ladder

M0 Qwen3-1.7B: full discovery/training matrix.

M1 Qwen3-4B: reduced qualification.

M2 Qwen3-8B: reduced qualification.

M3 Llama-3.1-8B: cross-family qualification.

13 Baselines

Mandatory baselines:

1. packed RAG;
2. independent pre-RoPE PRA;

3. Paper 3.2 rank-8 residual;
4. fixed sparse boundary mask;
5. oracle sparse interaction;
6. learned sparse selector/composer.

14 Metrics

Task quality. Official score, token F1, exact match, gold-answer NLL, and answer log-probability.

Distributional fidelity. First-step JS/KL, logit differences, and generated-output agreement.

Sparsity. Selected document pairs, layers, heads, tokens, and total cross-document interaction edges.

Systems. Persistent K/V reuse, newly computed states, request-local FLOPs, composition latency, TTFT, total latency, and memory.

Robustness. Five-seed intervals, order sensitivity, model-scale transfer, family transfer, and precision sensitivity.

15 Inception Results

15.1 Scope and endpoint validity

We ran the complete oracle grid on ten deterministically selected MultiHop-RAG questions with Qwen3-1.7B-4bit on an Apple M4 Pro with 48 GiB unified memory. BM25 produced 20 candidates, BGE-v2-M3 reranked them, and a 512-token budget retained three or four records. The model, reranker, corpus, and dataset revisions are pinned in the manifest. This is a one-seed mechanism cohort, not a population estimate. A post-freeze provenance check places all ten identities in the Paper 3.3 training partition, with no overlap with the Paper 3.2 final evaluation identities. The validation diagnostic and powered test confirmation therefore remain disjoint from this inception result.

Observer instrumentation matched ordinary host execution on all 10 questions. The 100% physical-edge plan also matched the instrumented packed teacher on all 10, with zero K/V delta and zero first-step JS. The 0% plan matched the packed no-cross-document mask. The average teacher graph contained 38.04 million physical cross-document edges over 28 layers and 16 heads. These checks close the implementation gate: differences inside the sweep are caused by the edge mask, not by an alternate attention kernel.

15.2 Quality–sparsity frontier

At 0.1% physical edges, the oracle recovers 97.3% of this ten-question cohort’s small 0.0059 F1 packed-independent gap: F1 is .1488 versus .1490 for packed RAG and .1431 for independent PRA. Official score remained .600 in all three conditions, so no official-score gap existed to recover. The 0.1% plan did not reproduce the same strings as packed RAG: output agreement was 0/10 and

Table 2: Paper 3.3 oracle smoke. “Edges” is the fraction of physical layer/head/token-pair cross-document edges. Attention mass is measured under the packed teacher.

Condition	F1	Official	Gold NLL	Edges	Mass
Packed host / oracle 100%	.1490	.600	4.780	100%	100%
Independent PRA	.1431	.600	2.550	0%	0%
Oracle top 0.01%	.1449	.600	2.579	.01%	4.0%
Oracle top 0.05%	.1416	.600	2.571	.05%	19.2%
Oracle top 0.1%	.1488	.600	2.610	.1%	36.2%
Oracle top 0.5%	.0968	.400	4.794	.5%	80.3%
Oracle top 5%	.0990	.400	4.851	5%	91.9%
Oracle top 10%	.1212	.500	4.852	10%	95.1%
Mass oracle 50%	.1403	.500	2.753	.149%	50%
Mass oracle 75%	.1145	.500	3.757	.324%	75%
Mass oracle 90%	.0768	.400	4.758	3.466%	90%
Mass oracle 99%	.1308	.500	4.855	30.902%	99%

first-step JS remained .110. The result is therefore task-level headroom, not behavioral equivalence, and the relative percentage should not be read as a large absolute quality gain.

The frontier is strongly non-monotonic. Increasing the top-edge budget from 0.1% to 0.5% reduced F1 from .1488 to .0968 even though retained teacher mass rose from 36.2% to 80.3%. Likewise, the smallest set carrying 50% of teacher mass used only .149% of physical edges but did not outperform independent PRA. Raw attention mass is thus an inadequate causal-utility target by itself.

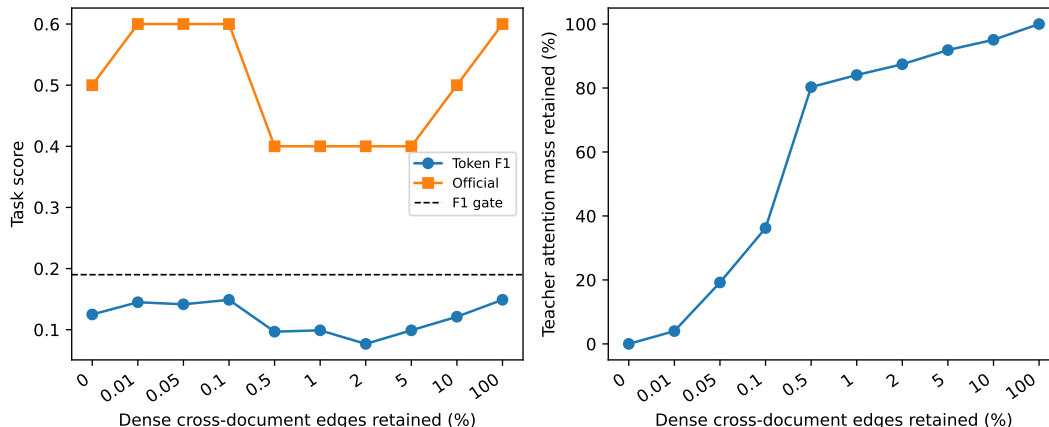


Figure 2: Task quality and retained teacher mass across physical top-edge budgets. The curve is a mechanism smoke over ten questions; connecting lines do not imply monotonicity or statistical precision.

15.3 Interaction localization

Cross-document attention mass was distributed across the network: layers 0–8 contained 27.4%, layers 9–18 contained 31.7%, and layers 19–27 contained 40.8%. No single layer exceeded 4.85% on average, and the largest layer/head cell averaged only .357%. The graph is therefore late-leaning but

not localized to one layer or head. Pair mass was more concentrated: the first record supplied 49.1% of mass into the second and 38.8% into the third. These values partly reflect causal geometry and token counts; they are routing features to test, not evidence that early-ranked records are causally sufficient.

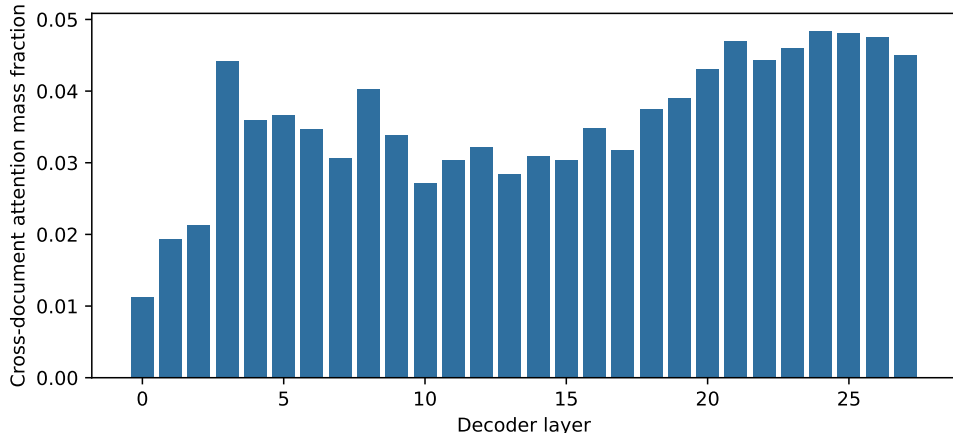


Figure 3: Mean share of packed-teacher cross-document attention mass by decoder layer. Later layers carry more aggregate mass, but no narrow layer band dominates.

15.4 Interventional validation

We next ran the six ranking targets on ten frozen validation questions, using the same model, selector, 512-token budget, and physical-edge replay. The BGE reranker ran on MPS; this execution device is part of the run configuration and receipt. All ten ordinary/instrumented host comparisons and all ten 100% replay checks passed exactly.

Table 3: Frozen validation target selection. Each oracle row reports its highest-F1 budget at or below 1%; it is not a test-set estimate.

Condition	Edges	F1	Official	Gold NLL
Packed teacher	100%	.1094	.600	5.911
Independent PRA	0%	.1673	.500	3.211
Raw attention	.1%	.1706	.500	3.697
Pair NLL	1%	.1659	.600	3.929
Pair JS	.5%	.1727	.600	3.666
Pair NLL $\times$ attention	.1%	.1379	.400	3.861
Layer NLL	.01%	.1540	.400	3.204
Layer JS	.5%	.1525	.500	3.739

This cohort reverses the inception endpoint: independent PRA exceeds packed F1 by .0579, although packed retains .100 more official score. It therefore does not define a packed advantage that sparse edges can meaningfully recover. The apparent improvements are also uncertain. Relative to the packed teacher, pair-JS at .5% changes F1 by +.0633 (paired bootstrap 95% interval $[-.0067, .1618]$) with zero mean official-score change; raw attention at .1% changes F1 by +.0613 ($[-.0275, .1604]$) while changing official score by $-.100$. Both intervals include zero. Gold NLL ranks

conditions differently from generated-answer metrics, so lower teacher-forced loss is not treated as proof of better task behavior.

No ranking has a non-decreasing F1 frontier through 1%. The result therefore strengthens, rather than resolves, the inception warning: raw magnitude is not a causal utility signal, and coarse leave-one-group-out NLL or first-step JS is not sufficient by itself. The powered test carries raw attention, pair NLL, pair JS, and pair-NLL-times-attention forward. Layer-only policies are dropped after validation because they do not improve the quality frontier.

15.5 Powered frozen-test confirmation

We then evaluate all 150 previously untouched test identities. The model is Qwen3-1.7B-4bit on an Apple M5 with 16 GiB unified memory; the BGE reranker runs on MPS. All 150 ordinary/instrumented comparisons and all 150 full-replay comparisons pass exactly. The run takes 18,486 seconds and records 4,800 condition rows. Table 4 reports the highest-F1 point at or below 1% for each ranking. These are test-set summaries selected over a small prespecified budget grid, not independently tuned deployment settings.

Table 4: Powered 150-question frozen test. Oracle rows show the best F1 budget at or below 1% physical cross-document edges.

Condition	Edges	F1	Official	Gold NLL
Packed teacher	100%	.1119	.3333	5.526
No-cross-document packed	0%	.1105	.3600	3.097
Independent PRA	0%	.1129	.3667	3.114
Raw attention	.05%	.1148	.3867	3.100
Pair JS	1%	.1205	.3800	3.817
Pair NLL	1%	.1243	.3933	3.675
Pair NLL $\times$ attention	.1%	.1215	.4000	3.319

Pair NLL at 1% changes F1 by +.0124 relative to packed RAG (paired bootstrap 95% interval $[-.0055, .0315]$) and official score by +.0600 ($[-.0133, .1333]$). The combined proxy at .1% changes F1 by +.0096 ($[-.0078, .0289]$) and official score by +.0667 ($[-.0067, .1400]$). Pair JS at 1% and raw attention at .05% also have intervals spanning zero. The powered result therefore distinguishes mean frontier shape from established superiority: causal grouping produces the better sparse means, but no task-quality effect is conclusive.

The endpoint ordering is itself informative. Packed F1 differs from independent PRA by only $-.0010$, and its paired interval includes zero. Gold NLL strongly prefers the blocked and independent conditions, so NLL divergence from packed does not translate monotonically into generated-answer quality. This cohort cannot support a claim that dense packed contextualization is a better task endpoint than a sparse oracle merely needs to recover.

The powered .1% plans also reveal different geometries. Raw attention assigns 32.6% of edges to layers 23–25 and 27 and 94.3% to pairs (0, 1) and (0, 2). Pair NLL spreads 84.8% across (0, 2), (1, 2), and (0, 1) and emphasizes early layers 1–2 alongside distributed later layers. Pair JS assigns 86.7% to (0, 1), while the combined proxy assigns 91.7% across the first three pairs and leans later. No leading layer–head cell receives more than .86% of a policy’s selected edges. These are policy-induced concentrations, not universal heads, but they show that the causal target materially changes the proposed sparse computation.

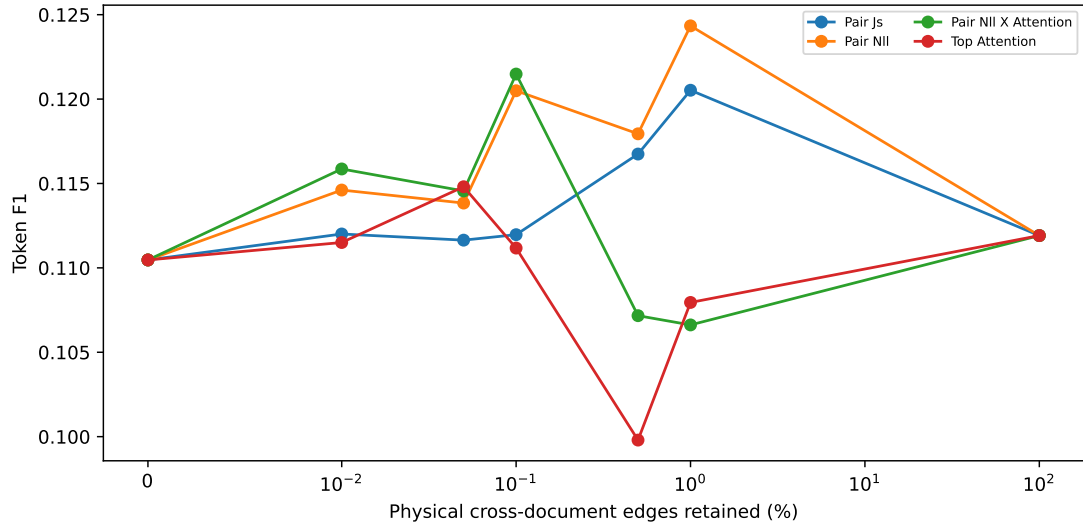


Figure 4: Powered-test F1 frontiers for observational and interventional rankings. Causal pair rankings improve the mean sparse frontier, but all curves remain non-monotonic; lines aid reading and do not imply interpolation.

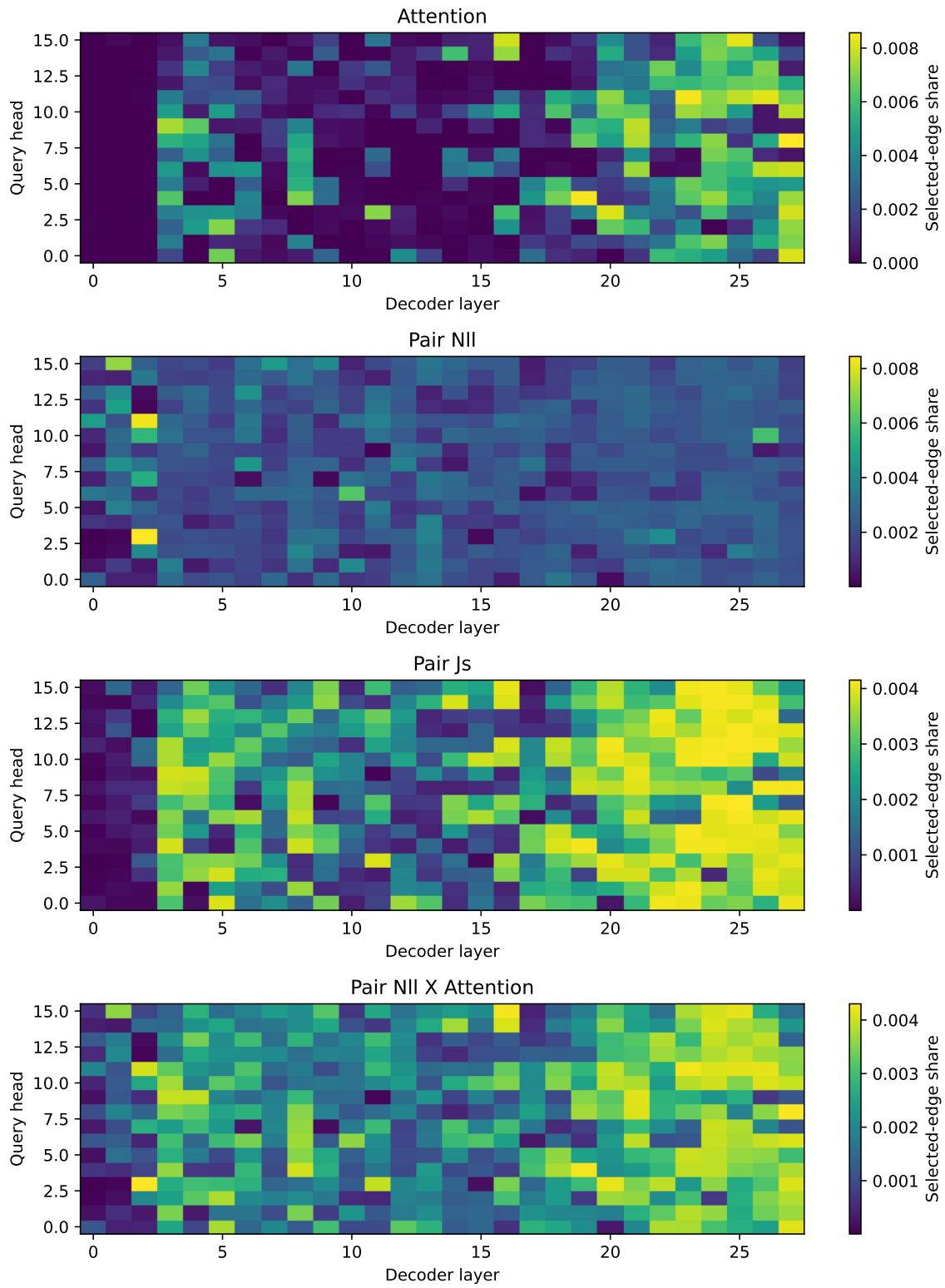


Figure 5: Layer-head allocation of each .1% plan over 150 frozen test questions. Color is selected-edge share within a policy.

15.6 Go/no-go decision

The powered gate is **locked**. The cohort-size requirement is met, but no condition reaches .19 F1 and .67 official score, no frontier is non-decreasing through 1%, and the paired task-quality intervals include zero. Learned selector training therefore remains disabled. The next useful selector target must improve causal localization beyond one scalar utility per document pair; training the current target would turn an unresolved oracle into a learned approximation of an unstable policy.

The reported encoding times, roughly 0.5–0.9 seconds per condition, execute a dense host attention kernel under a sparse mask. Edge fraction is therefore a logical interaction budget, not a measured wall-time or FLOP reduction. A production sparse kernel is downstream work and should be evaluated only after the quality gate passes.

16 Success Criteria

The predeclared practical gate is

$$F1 \geq .19, \quad \text{official} \geq .67,$$

or statistical indistinguishability from packed RAG, while using no more than 5% of dense request-specific cross-document interaction. Approximate 80% gap recovery remains a useful descriptive quantity, but does not replace these absolute thresholds.

Neither the ten-question smoke nor the 150-question frozen test meets these absolute criteria. The smoke’s 0.1% point meets the relative F1-recovery target only because its packed and independent endpoints are close. On the powered test, causal rankings improve sparse mean quality but do not establish a monotonic frontier or a nonzero paired task-quality effect. We do not substitute either post-hoc observation for the prespecified gate.

A stronger result matches packed RAG within uncertainty while preserving:

- persistent native K/V reuse;
- lower order sensitivity;
- materially lower request-specific compute.

17 Falsification

Scientifically useful negative outcomes include:

- an oracle requires dense interaction;
- learned selection fails to approach the oracle;
- sparse state propagation destroys quality;
- selector overhead erases saved prefill;
- order robustness disappears at matched quality;
- the mechanism does not transfer across models/families.

18 Systems Integration

Paper 3.3 reuses the canonical Paper 4.5/Paper 3.2 `ContextRecord` and storage metadata. The learned interaction policy is request-local and must not mutate persistent source records.

The target runtime abstraction is

`ContextRecord` $\rightarrow$ `SelectionReceipt` $\rightarrow$ `InteractionPlan` $\rightarrow$ `RequestLocalComposition`.

An `InteractionPlan` should record selected record pairs, token spans, layers/heads, budget, policy revision, and model/runtime compatibility.

19 Discussion

The 0.1% result suggests that conventional RAG may perform substantially redundant document-to-document computation. It does not yet establish that claim: the sample is small, the absolute gate failed, and the present sparse mask still executes a dense kernel. PRA could eventually retain persistent source state and pay only for a small request-specific neural “join” among selected resources, but that economic claim requires both powered quality and a real sparse execution path.

This would change the role of PRA from a cache/retrieval mechanism into a composable native-memory substrate:

persistent records + sparse request-local contextualization.

If it fails, the result would instead bound how separable transformer source context really is: packed causal contextualization may be too distributed to recover cheaply after independent encoding.

20 Conclusion

Paper 3.2 established that position transport and persistent native memory can be separated from the cross-document causal computation induced by packed RAG. Paper 3.3 targets that remaining computation directly. Exact endpoint checks show that the observer and physical-edge masks isolate this variable. The ten-question smoke finds a promising 0.1% point but also a sharply non-monotonic frontier, so attention mass alone is not a reliable selector target. The 150-question frozen test confirms all endpoints and finds that document-pair causal rankings improve mean sparse quality over raw magnitude: pair NLL reaches .1243 F1 at 1% edges versus .1119 packed. The paired F1 interval nevertheless includes zero, the packed endpoint is not stronger than independent PRA, every frontier remains non-monotonic, and the absolute gate is missed. Sparse causal headroom is credible; a qualified selector is not. Learned-selector training remains locked until a finer task-aware utility signal produces a stable frontier.

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A Interventional Replay Procedure

For each frozen selection, the implementation performs the following steps:

1. Encode the concatenated records with ordinary full-causal attention, observing only causal cross-record probabilities into a compressed $[L, H, |E|]$ graph.
2. Encode the same tokens with every cross-record edge blocked. This is the base mask to which sparse plans add visibility.
3. For each document-pair or layer group g , replay the full graph except g , score the gold answer without decoding, and record $u_{\text{NLL}}(g)$ and $u_{\text{JS}}(g)$.
4. Order groups by one measured utility. Within a group, order physical edges by teacher attention, or multiply non-negative group NLL utility by edge attention for the combined proxy.

5. Replay nested prefixes at each physical-edge budget. Generate and teacher-force the answer from the same immutable selection and query.
6. Verify that the empty plan matches the blocked control and that the full plan matches the instrumented packed teacher exactly.

The implementation obtains only the prefix needed by the largest sparse budget through deterministic partition selection. It does not sort all tens of millions of edges for each target. This optimization changes oracle-analysis latency, not the selected set or the dense masked host execution used to test quality.

B Artifact Schema

Each run should persist:

- dataset/query IDs and seed;
- candidate and selection receipts;
- persistent record URIs/versions;
- document order;
- teacher interaction graph digest;
- oracle/learned interaction-plan digest;
- leave-one-group-out NLL and first-step JS utility;
- selected pairs/layers/heads/tokens;
- interaction budget;
- training checkpoint/policy revision;
- precision/runtime metadata;
- task/distribution/system metrics;
- paired bootstrap effects against the packed teacher;
- reranker execution device and frozen-split digest;
- git commit and hardware.

C Reproduction Map

The physical graph, plan, and localization implementation is in `src/pra_hf/sparse_crossdoc.py`. The host-preserving observer and sparse mask seam is in `src/pra_hf/rag_mlx_native.py`. The complete frozen retrieval and oracle sweep is in `experiments/paper3_3_sparse_crossdoc/run_oracle_sparsity.py`; split generation is in the adjacent `freeze_splits.py`. Cross-record search, authorization, logical-interval deduplication, accounting, and qualification gates are implemented in `src/pra_hf/crossdoc_expansion.py`. Retrieval- linked pair and boundary plans are in `src/pra_hf/sparse_crossdoc.py`. The evidence frontier and native answer replay are respectively `experiments/paper3_3_crossdoc_expansion/run_expansion_frontier.py` and `experiments/paper3_3_crossdoc_expansion/run_expansion_generation.py`. Exact commands and evidence labels appear in the experiment and paper README files.

The tracked publication summary named in the README contains aggregate metrics, parity counts, monotonicity diagnostics, paired intervals, runtime metadata, localization summaries, and a SHA-256 digest of the full run manifest. The larger per-question manifest, regenerable teacher graphs, and recovery checkpoints remain on the experiment host rather than in Git. Group interventions and publication figures are tracked separately.